

PTVA's MULUND COLLEGE OF COMMERCE (AUTONOMOUS)		
Programme: B.Sc. Information Technology	Class: TY	Semester: V
Exam: CIA	Name of the Course: Artificial Intelligence	
Date of Exam: 30/07/2025	Marks: 20	Time: 40 Minutes
Instructions: 1) Make suitable assumptions wherever necessary and state the assumptions made. 2) Draw neat labelled diagrams wherever necessary.		

Q Attempt any four (5 marks each)

- Demonstrate your understanding of Artificial Intelligence by describing what it is and explaining any five of its practical applications.
- Define the following types of agents and give one example of each. (i) Rational agent; (ii) Irrational agent; (iii) Learning agent; (iv) Omniscient agent; (v) Autonomous agent.
- Examine the working of a model-based agent using a labelled diagram.
- Compare BFS and DFS and explain how they are different.
- Apply the 8-puzzle problem to the given scenario by defining the initial and goal states. Show the steps and calculate the path cost.

1	2	3
4	8	----
7	6	5

Initial State

1	2	3
4	5	6
7	8	----

Goal State

- Define the state-space search diagram used in the hill climbing algorithm and describe its basic structure.

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